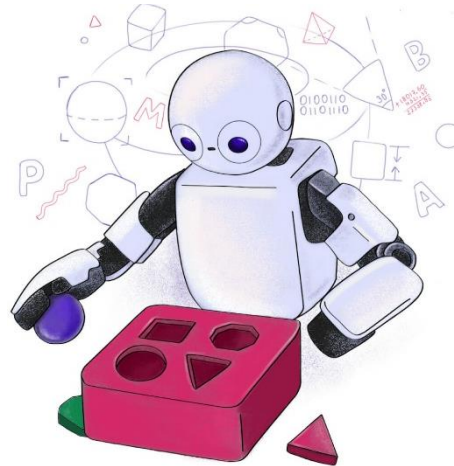
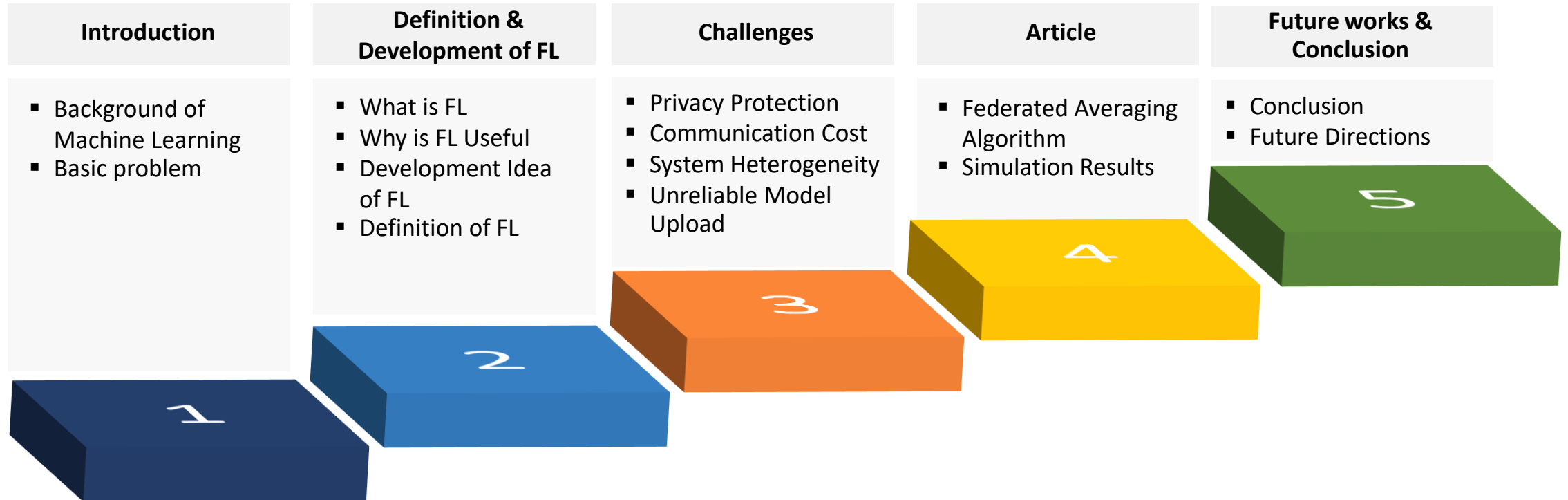


TP558 - Tópicos avançados em Machine Learning

Federated Learning



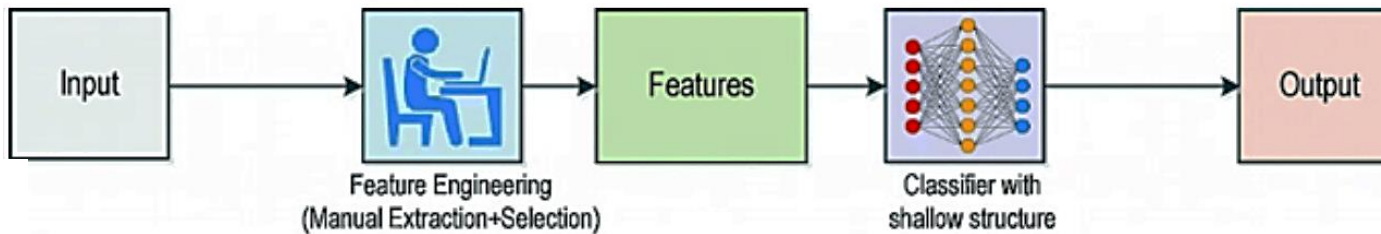
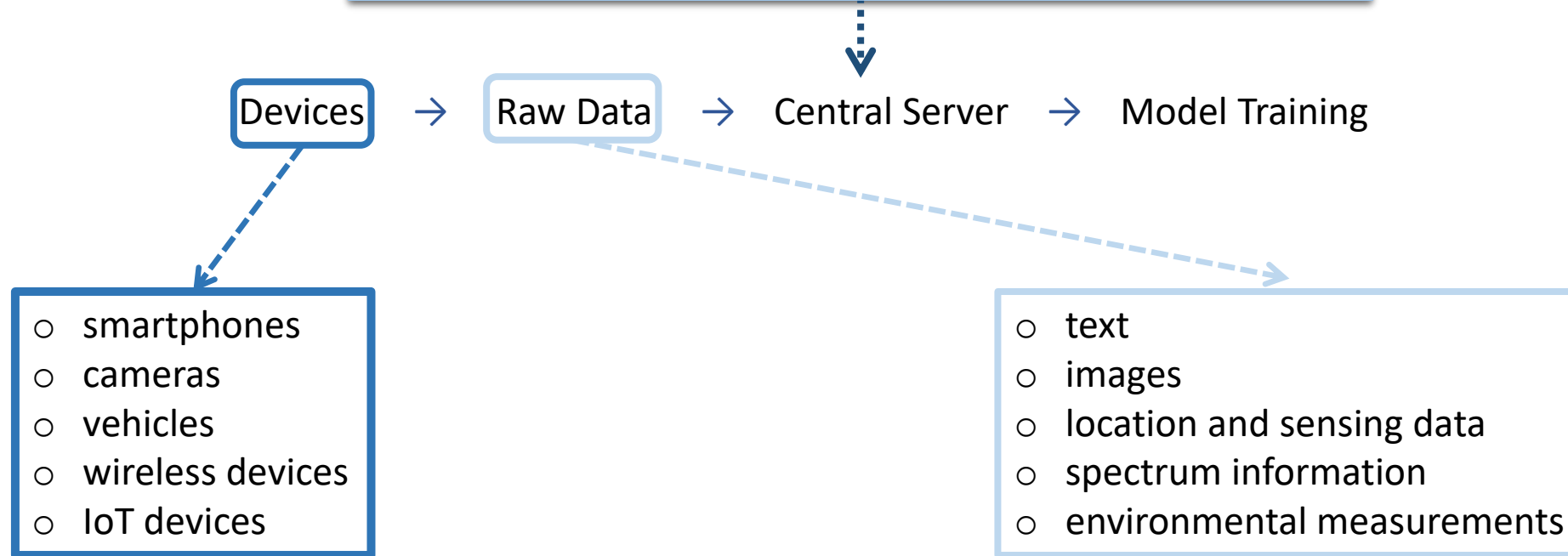
Outline

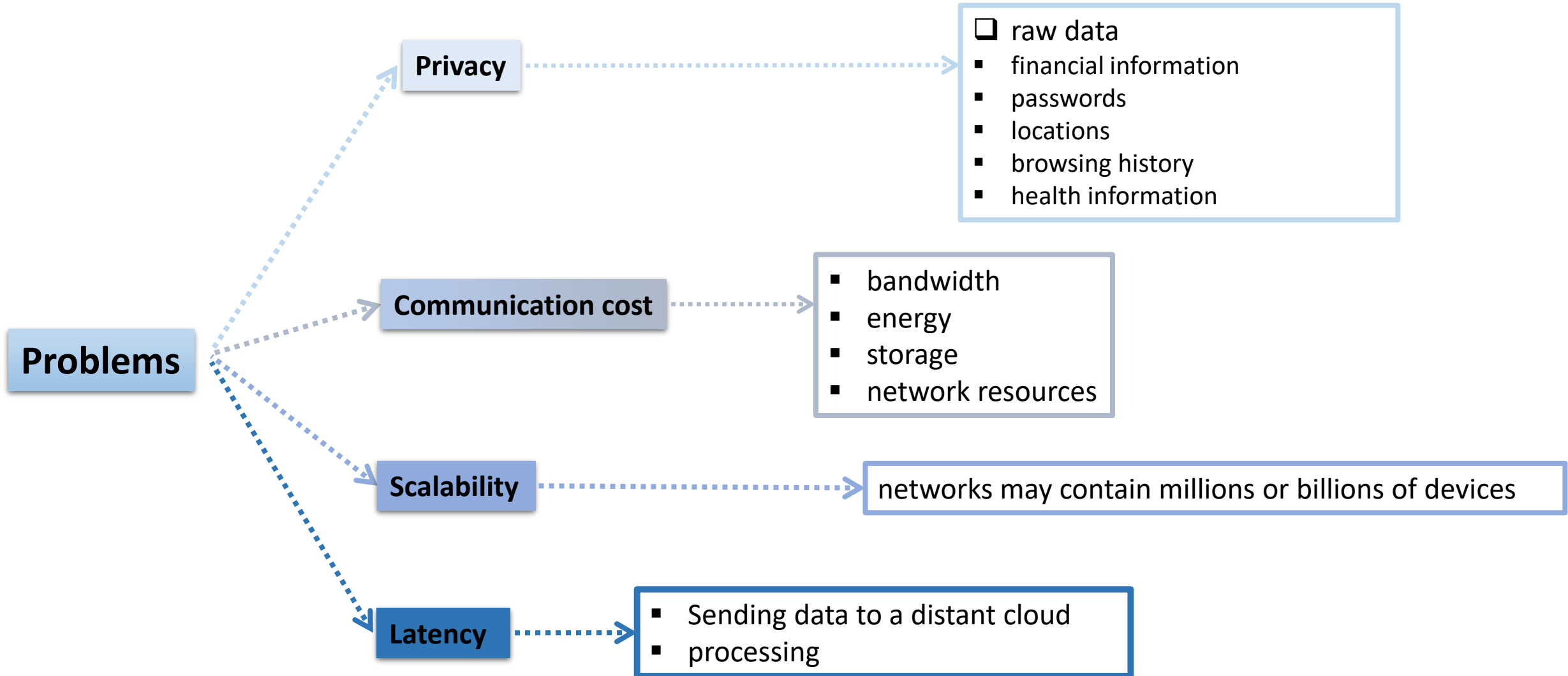


Introduction

Introduction

Traditional machine learning generally follows this process





Definition and Development of Federated Learning

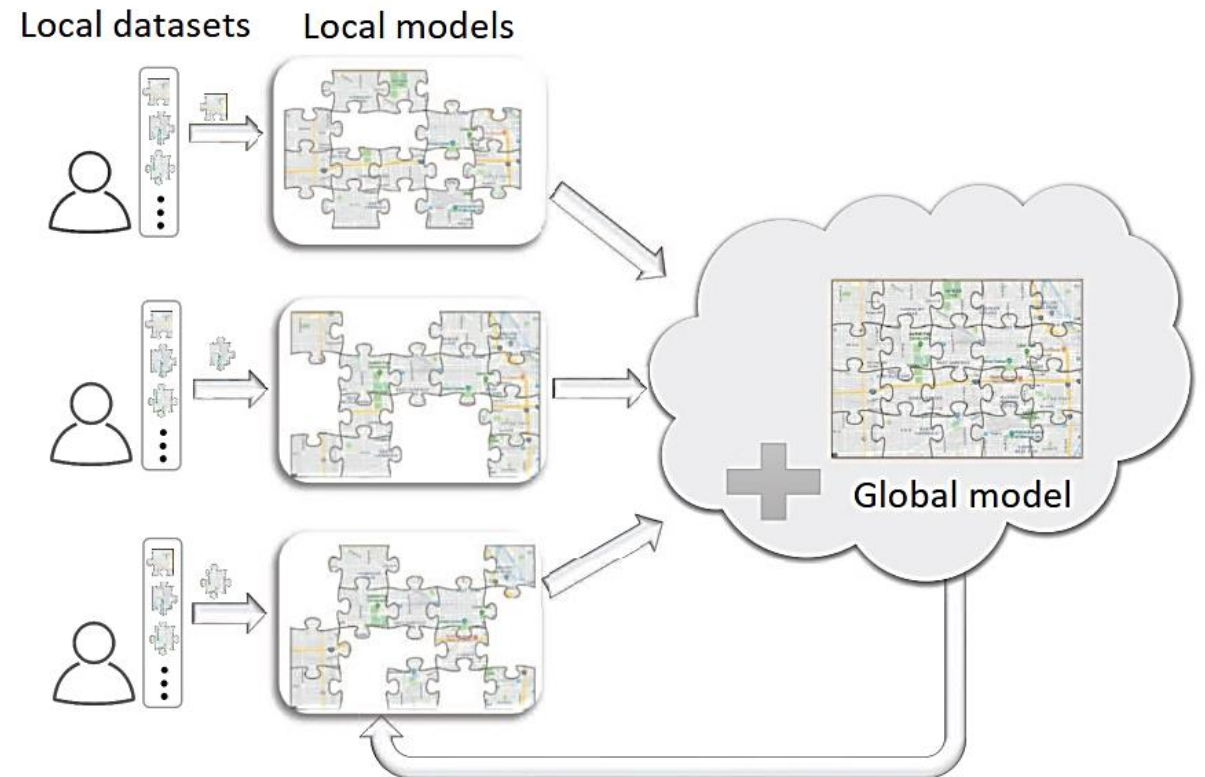
What Is Federated Learning?

1. Local computing

The data stays on the local device

2. Model transmission

Instead of sending raw data, clients exchange model-related information



Why is Federated Learning Useful?



Directly sending their privacy-sensitive data to a central server



Development Idea of Federated Learning

Several companies want to collaboratively train a model

Each company has

- limited data
- incomplete data
- privacy restrictions

❑ If each company trains independently

Model Quality ↓

Why



Each organization **only** sees part of the available information

❑ If they directly combine all data

Privacy Risk ↑

Federated learning tries to solve both problems

Collaboration + Data Privacy

Definition of Federated Learning and Federated Averaging

1. Each client performs local training

$$\omega_t \rightarrow \omega_{t+1}^{(n)}$$

2. The clients then send their updated models to the server

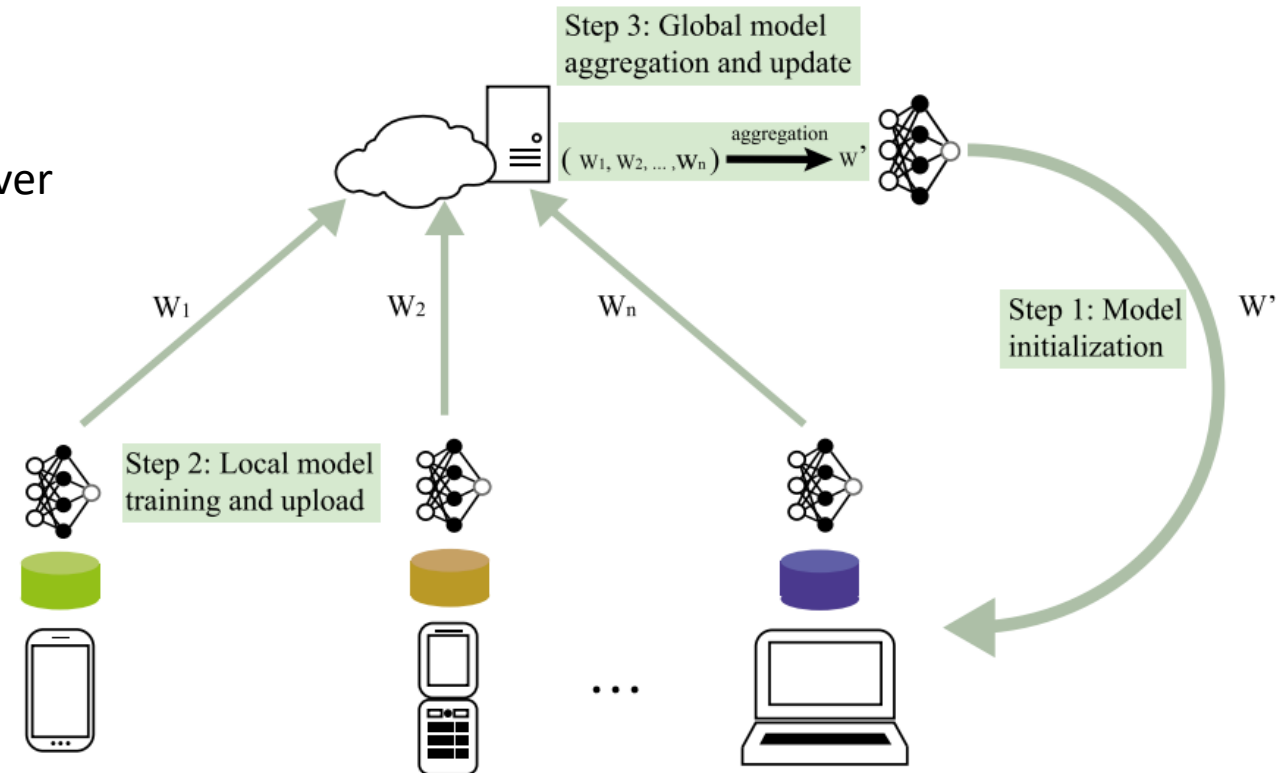
$$\omega_{t+1}^{(1)}, \omega_{t+1}^{(2)}, \dots, \omega_{t+1}^{(n)}$$

3. The server aggregates them

$$\omega_{t+1} = \sum_{n=1}^N \frac{m_n}{m} \omega_{t+1}^{(n)}$$

- n number of participating clients,
- m_k number of training samples at client k
- $m = \sum_n m_n$ total number of samples
- ω_n Local Model
- ω' global mod

4. The resulting global model is distributed again



Challenges

Challenges

❑ Privacy Protection

↑ Privacy ↔ Computational Cost ↑

❑ Communication Cost

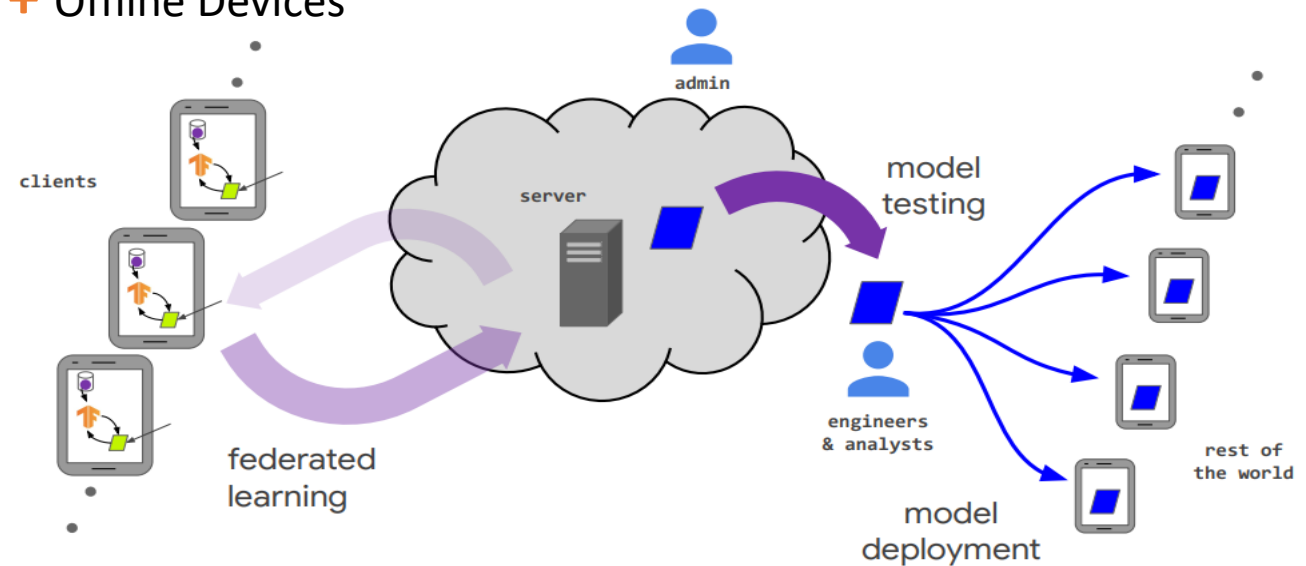
↑ Communication with Millions of Devices ↔ Communication Efficiency ↑

❑ System Heterogeneity

Different Hardware + Different Connections + Offline Devices

❑ Unreliable Model Upload

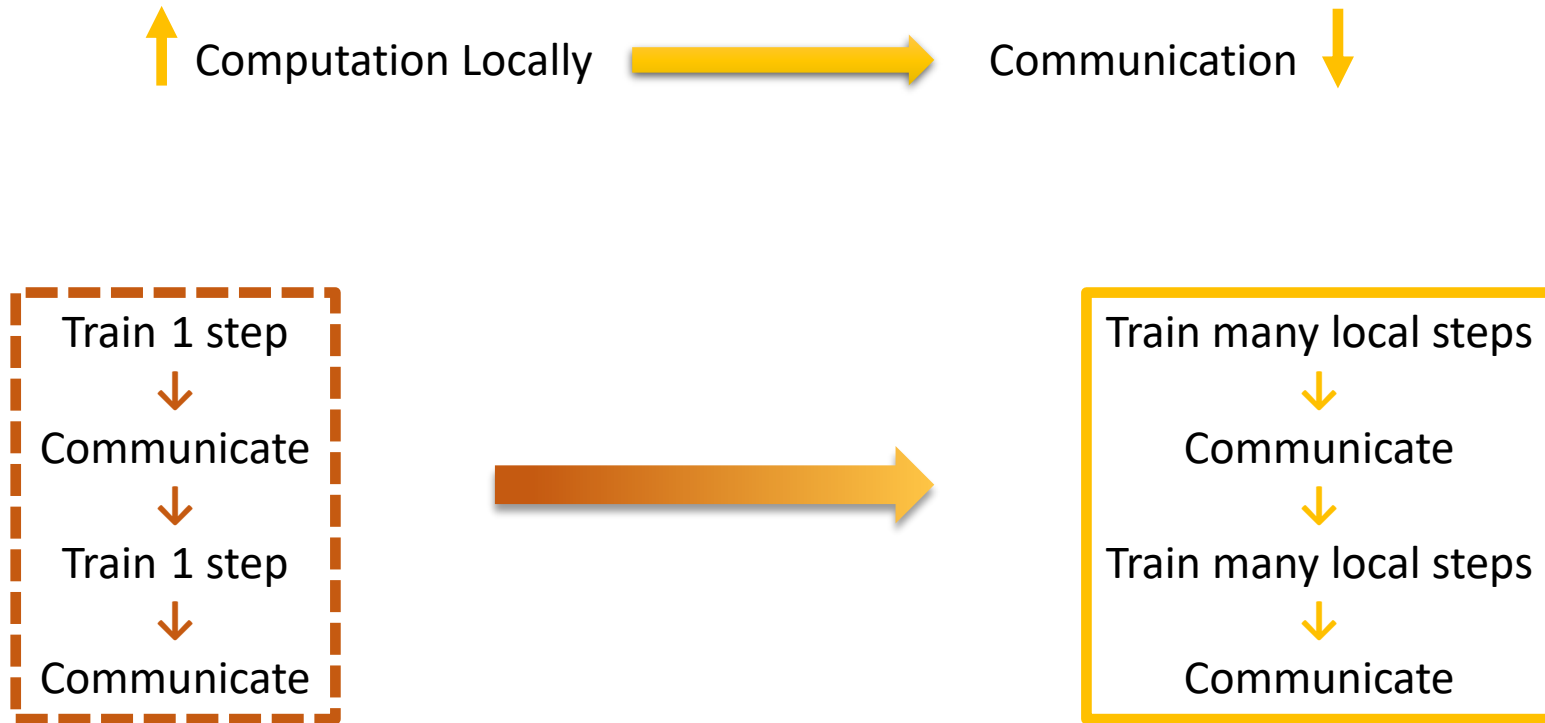
- Malicious Attack
- Unstable Network



Article

Communication-Efficient Learning of Deep Networks from Decentralized Data

Hypothesis



Brendan McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data" arXiv, 2023. doi: 10.48550/arXiv.1602.05629.

Contributions

✓ Contribution 1

Identifying decentralized learning from mobile devices as an important research direction

✓ Contribution 2

Introducing **Federated Averaging (FedAvg)**

✓ Contribution 3

Showing robustness to

- unbalanced data
- non-IID data

and dramatic reductions in communication rounds

Brendan McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data" arXiv, 2023. doi: 10.48550/arXiv.1602.05629.

Mathematical Formulation

$$\min_{\omega \in \mathbb{R}^d} f(\omega) \rightarrow f(\omega) = \frac{1}{n} \sum_{i=1}^n f_i(\omega)$$

- ω sretemarap krowten laruen =
- n elpmaxe gniniart fo rebmun =
- $f_i(\omega)$ i elpmaxe rof ssol =

Dividing the dataset across k clients

$$n_k = |P_k|$$

number of examples at client k

set of data points stored at client k

Local objective

$$F_k(w) = \frac{1}{n_k} \sum_{i \in P_k} f_i(w)$$

Global objective

$$f(w) = \sum_{k=1}^K \frac{n_k}{n} F_k(w)$$

Brendan McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data" arXiv, 2023. doi: 10.48550/arXiv.1602.05629.

IID vs Non-IID Data

IID $\longrightarrow$ local data is approximately representative of the global distribution

IID client have approximately the same proportions $\longrightarrow F_k(w) \approx f(w)$

Non-IID $\longrightarrow F_k(w) \not\approx f(w)$

Why is this difficult?

client can optimize perfectly for its own data while moving the model in a direction that is not globally optimal

Brendan McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data" arXiv, 2023. doi: 10.48550/arXiv.1602.05629.

Communication vs Computation

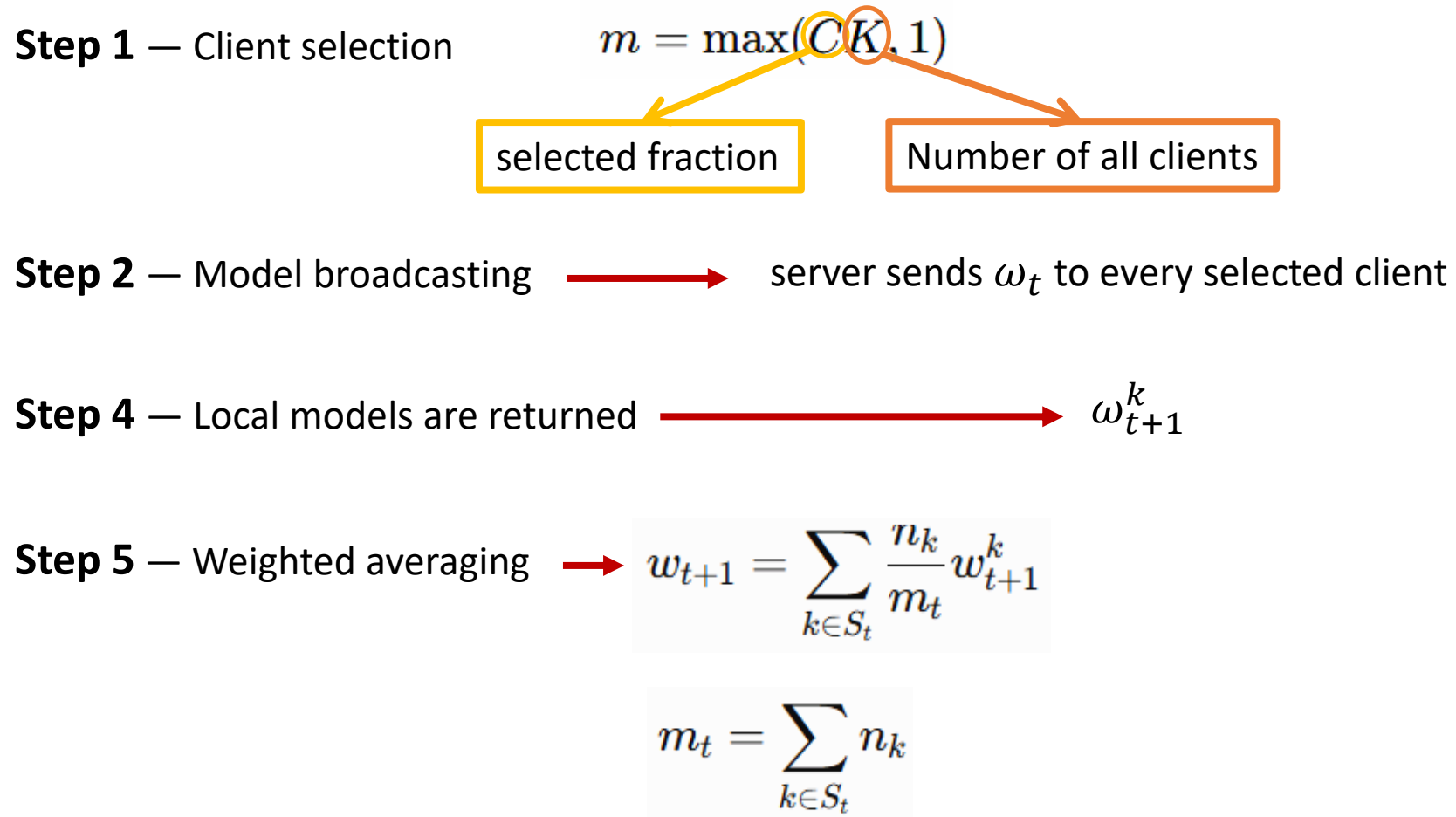
Goal  Use more local computation  Reduce communication rounds

Two ways to increase computation

- **More parallelism**
Use more clients in each round.
- **More local computation**
Each participating client performs more SGD updates before communicating.

Brendan McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data" arXiv, 2023. doi: 10.48550/arXiv.1602.05629.

FedAvg Workflow



Brendan McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data" arXiv, 2023. doi: 10.48550/arXiv.1602.05629.

Algorithm 1 FederatedAveraging. The K clients are indexed by k ; B is the local minibatch size, E is the number of local epochs, and η is the learning rate.

Server executes:

```

initialize  $w_0$ 
for each round  $t = 1, 2, \dots$  do
   $m \leftarrow \max(C \cdot K, 1)$ 
   $S_t \leftarrow$  (random set of  $m$  clients)
  for each client  $k \in S_t$  in parallel do
     $w_{t+1}^k \leftarrow \text{ClientUpdate}(k, w_t)$ 
   $m_t \leftarrow \sum_{k \in S_t} n_k$ 
   $w_{t+1} \leftarrow \sum_{k \in S_t} \frac{n_k}{m_t} w_{t+1}^k$  // Erratum4

```

```

ClientUpdate( $k, w$ ): // Run on client  $k$ 
   $\mathcal{B} \leftarrow$  (split  $\mathcal{P}_k$  into batches of size  $B$ )
  for each local epoch  $i$  from 1 to  $E$  do
    for batch  $b \in \mathcal{B}$  do
       $w \leftarrow w - \eta \nabla \ell(w; b)$ 
  return  $w$  to server

```

C — Client fraction (percentage of clients participate in each round)

E — Local epochs (number of complete passes over the client's local dataset)

B — Local minibatch size (Instead of training on the complete local dataset at once, divide it into minibatches)

η — Learning rate (controls the step size of local SGD)

Effect of Local Computation

□ Number of local updates per client per round $\longrightarrow u = \frac{En}{KB}$

- E = local epochs
- n = total number of examples
- K = number of clients
- B = minibatch size

□ More local computation $\Rightarrow$ Fewer communication rounds

$E \uparrow \Rightarrow \text{communication} \downarrow$

$E \uparrow \uparrow \Rightarrow \text{possible instability}$

□ Very large numbers of local epochs

- plateauing
- divergence

MNIST CNN, 99% ACCURACY					
CNN	E	B	u	IID	Non-IID
FEDSGD	1	∞	1	626	483
FEDAVG	5	∞	5	179 (3.5 $\times$)	1000 (0.5 $\times$)
FEDAVG	1	50	12	65 (9.6 $\times$)	600 (0.8 $\times$)
FEDAVG	20	∞	20	234 (2.7 $\times$)	672 (0.7 $\times$)
FEDAVG	1	10	60	34 (18.4 $\times$)	350 (1.4 $\times$)
FEDAVG	5	50	60	29 (21.6 $\times$)	334 (1.4 $\times$)
FEDAVG	20	50	240	32 (19.6 $\times$)	426 (1.1 $\times$)
FEDAVG	5	10	300	20 (31.3 $\times$)	229 (2.1 $\times$)
FEDAVG	20	10	1200	18 (34.8 $\times$)	173 (2.8 $\times$)

SHAKESPEARE LSTM, 54% ACCURACY					
LSTM	E	B	u	IID	Non-IID
FEDSGD	1	∞	1.0	2488	3906
FEDAVG	1	50	1.5	1635 (1.5 $\times$)	549 (7.1 $\times$)
FEDAVG	5	∞	5.0	613 (4.1 $\times$)	597 (6.5 $\times$)
FEDAVG	1	10	7.4	460 (5.4 $\times$)	164 (23.8 $\times$)
FEDAVG	5	50	7.4	401 (6.2 $\times$)	152 (25.7 $\times$)
FEDAVG	5	10	37.1	192 (13.0 $\times$)	41 (95.3 $\times$)

Brendan McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data" arXiv, 2023. doi: 10.48550/arXiv.1602.05629.

23× Communication Reduction Result

FedSGD

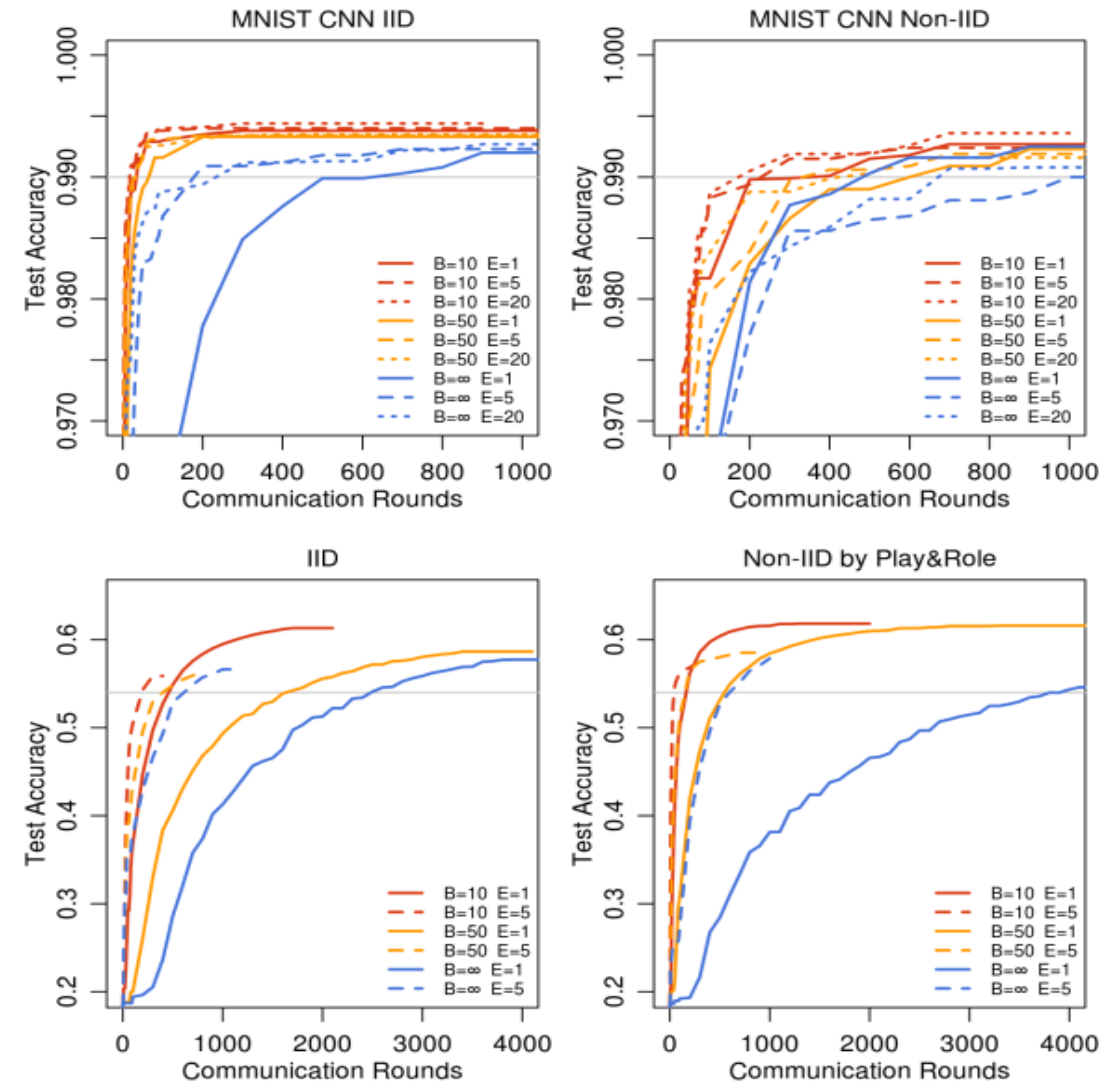
Required 820 communication rounds to achieve 10.5% test accuracy.

FedAvg

Required only 35 communication rounds to achieve the same 10.5% test accuracy.

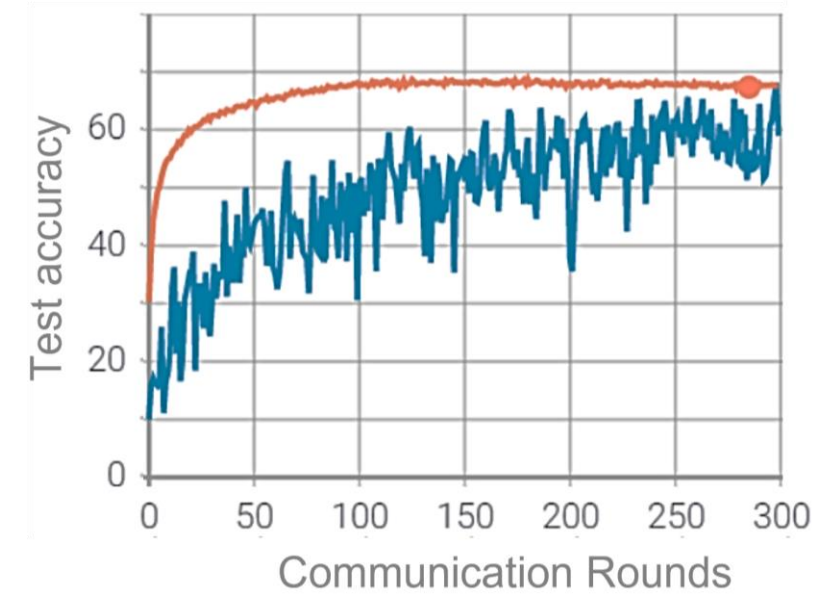
$$820 / 35 \approx 23.4$$

approximately 23× fewer communication rounds



Brendan McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data" arXiv, 2023. doi: 10.48550/arXiv.1602.05629.

	CIFAR 10	MNIST
IID	68.70 %	98.87 %
Non-IID	68.11 %	98.05 %



Model CNN, Dataset CIFAR or MNIST, epochs 100 and 300, η 0.01, local-ep 5, local bs 32

Future works and Conclusion

Strengths

- Simplicity
- Communication efficiency
- Non-IID robustness
- Model diversity
- Practical motivation

Limitations

- Privacy is not guaranteed
- Client heterogeneity
- Synchronous operation
- Non-IID remains difficult
- Hyperparameter selection

Future Directions

1. Privacy

make the training process mathematically private

2. Security

defend against

- malicious clients
- poisoned updates
- model inversion
- inference attacks

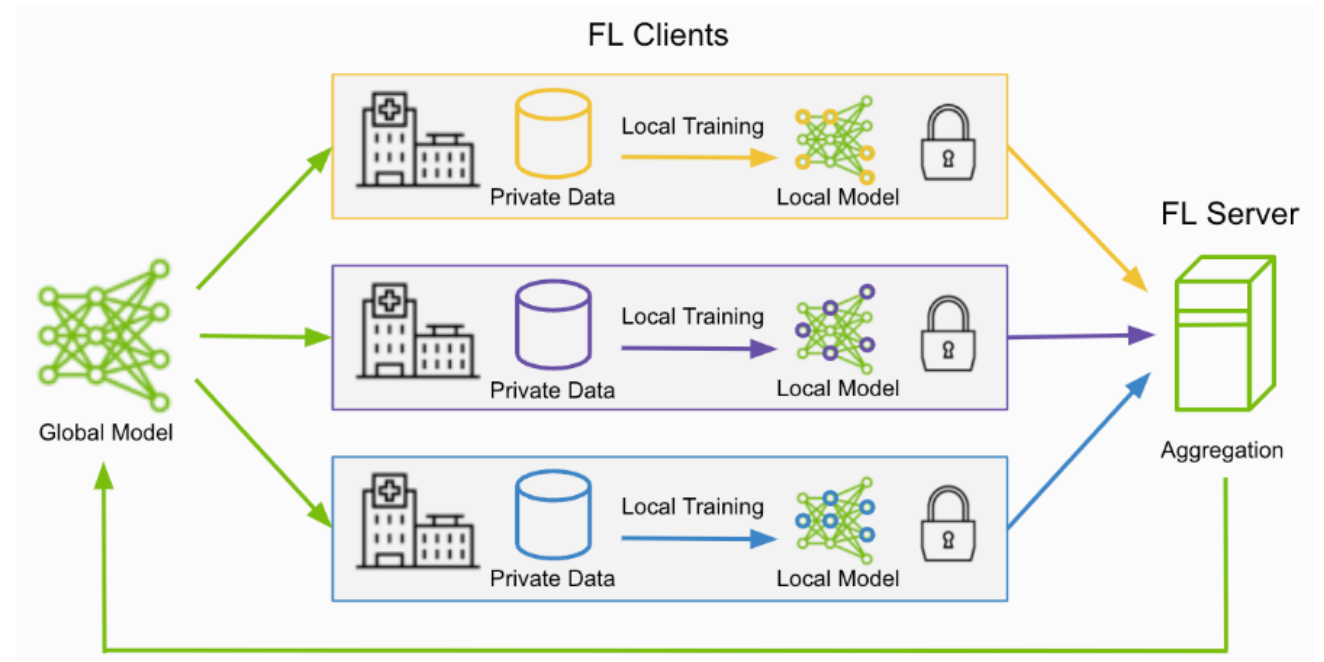
3. Communication

reduce

- model size
- uplink traffic
- downlink traffic
- number of rounds

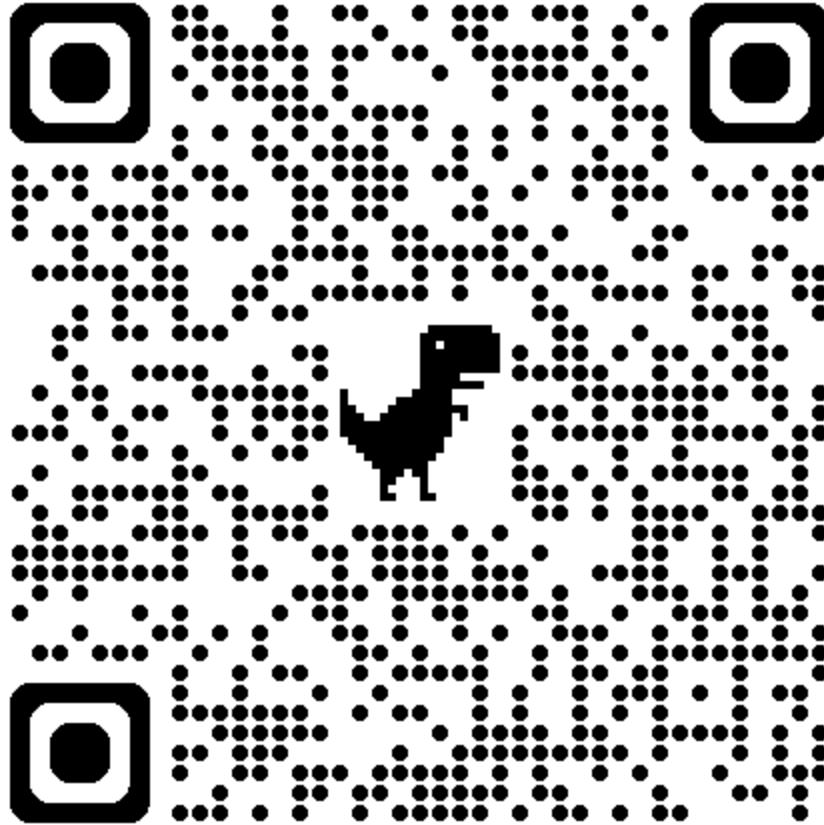
4. Heterogeneity

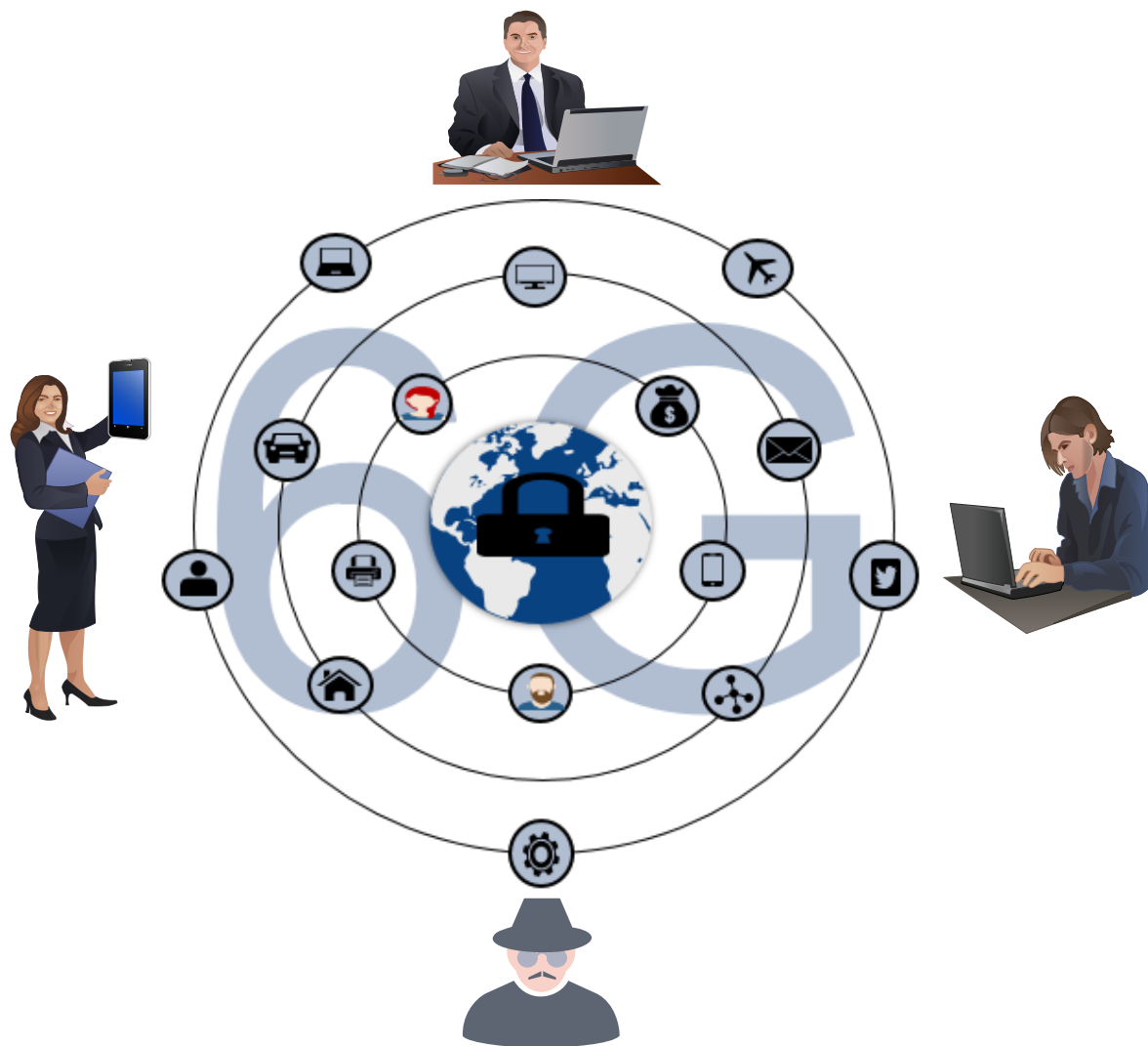
- different device capabilities
- different network conditions
- different local datasets



- ❑ S. Niknam, H. S. Dhillon and J. H. Reed, "Federated Learning for Wireless Communications: Motivation, Opportunities, and Challenges," in *IEEE Communications Magazine*, vol. 58, no. 6, pp. 46-51, June 2020, doi: 10.1109/MCOM.001.1900461.
- ❑ Chen Zhang, Yu Xie, Hang Bai, Bin Yu, Weihong Li, Yuan Gao, "A survey on federated," in *sciencedirect*, Volume 216, 2021, ISSN 0950-7051, doi: 10.1016/j.knosys.2021.106775.
- ❑ Peter Kairouz, et al., "Advances and Open Problems in Federated Learning," *arXiv*, 2021, doi: 10.48550/arXiv.1912.04977.
- ❑ Brendan McMahan et al., "Communication-Efficient Learning of Deep Networks from Decentralized Data" *arXiv*, 2023, doi: 10.48550/arXiv.1602.05629.

Quiz





Thank you for your
attention